Experiment 9: Process Lifecycle, Hierarchy & Diagnostic Monitoring

Source file: process_info.c **Type:** C program

Source Code

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
int main(void) {
   printf("Process ID (PID): %ld\n", (long)getpid());
   printf("Parent Process ID (PPID): %ld\n", (long)getppid());
   printf("User ID: %ld\n", (long)getuid());
   printf("Process will pause for 5 seconds...\n");
   sleep(5);
   puts("Process completed.");
   return 0;
}
```

Output :

```
safa@DESKTOP-SVKNP0S:~$    vi process_info.c
safa@DESKTOP-SVKNP0S:~$    gcc -Wall -Wextra process_info.c -o process_info
safa@DESKTOP-SVKNP0S:~$    ./process_info
Process ID (PID): 859
Parent Process ID (PPID): 309
User ID: 1000
Process will pause for 5 seconds...
Process completed.
safa@DESKTOP-SVKNP0S:~$    ps -o pid,ppid,stat,comm -C process_info
    PID    PPID STAT COMMAND
safa@DESKTOP-SVKNP0S:~$
```